

Altera University — Hardware Donation Request

Form Entries

Page 1 — Donation Request

Field	Entry
Intended Use	Research
Boards Selection	DE10-Agilex
Quantity	2
Use Address on your Profile to ship?	No
Shipping Country/Territory	Italy
Shipping City	Palermo
Shipping State/Province	Sicilia (PA)
Shipping Address	CNR — Istituto di Calcolo e Reti ad Alte Prestazioni (ICAR), Palermo

Applicant

Field	Entry
Name (PI / institutional contact)	Prof. Gianpaolo Vitale
Institution	CNR — Istituto di Calcolo e Reti ad Alte Prestazioni (ICAR)
Institution type	Not-for-profit research institute
ORCID	0000-0002-5439-6835
Lead researcher	Gianluigi Vitale — Computer Engineering student, Universitas Mercatorum, Rome; sole author of DriftBench (MLSys 2026)
Lead researcher ORCID	0009-0007-6740-2695

Note on hosting and relationship

The PI and the lead researcher are family; this is disclosed for transparency, and the institution's conflict-of-interest process will be followed. The hardware is hosted at **CNR-ICAR Palermo**, the PI's institution: the PI provides the institutional home, the physical lab space, and responsibility for the boards — import/customs handling, custody, and research supervision — consistent with the program's terms that donated boards are the property of the institution, not an individual. The lead researcher conducts the FPGA research at CNR-ICAR under that institutional umbrella. (Universitas Mercatorum, the lead

researcher's degree-granting university, is fully online and is not the hardware site; CNR-ICAR is the physical research location.)

Page 2 — Research

Research Title

Cross-Architecture Infrastructure Drift: Extending GPU-Validated Drift Measurement to FPGA-Based AI Inference

Research Objectives

DriftBench (G. Vitale, MLSys 2026, oral; <https://openreview.net/forum?id=Xfzzp6grRP>; dataset DOI 10.5281/zenodo.19361066) established that infrastructure drift – output and safety-verdict changes in a fixed model caused only by changing the hardware, precision, or serving framework underneath it – is measurable and predictable within a GPU family (Portability Risk Index, held-out $R^2 = 0.909$ over 236,985 prompt-response pairs and 105 configurations). The open question is whether that predictability transfers across fundamentally different accelerator architectures.

This project answers that question using FPGAs as the cleanest "very different architecture": spatial dataflow rather than SIMT, a different memory hierarchy, and a deterministic, statically-scheduled compute path whose numerical sequence is fixed at compile time.

Research questions:

1. Does systematic hardware drift, predictable within GPU families, generalize to the FPGA execution model, or does the different architecture create idiosyncratic patterns that require fresh measurement?
2. Can the Portability Risk Index predict GPU-to-FPGA migration drift, or does an accelerator-family change constitute a new drift category?
3. Does a deterministic, statically-scheduled FPGA compute path reduce or eliminate the output variation observed on GPUs, bracketing the drift phenomenon from the deterministic end?

Scope: the FPGA evaluation runs a small classifier model selected for compatibility with the FPGA AI Suite's supported layer set, deployed via the official DE10-Agilex PCIe design example. Measuring whether this model drifts on a deterministic spatial architecture – and characterizing how a statically-scheduled compute path bounds output variation relative to GPU runtime non-determinism – is a self-contained cross-architecture drift result.

Hardware: two DE10-Agilex boards enable an FPGA-to-FPGA variance study (whether identical boards produce bit-identical outputs, itself a potential drift source) while running experiments and development in parallel over a multi-month campaign.

Deliverables: an extended public DriftBench dataset with FPGA measurements (first

release within the delivery window), followed by an updated Portability Risk Index with FPGA support and deployment guidelines for GPU/FPGA infrastructure decisions. The full results target an MLSys 2027 submission. All code and data release under Apache 2.0 / MIT. Altera and Terasic will be acknowledged in the experimental-platform section of resulting publications.

Delivery date

First public dataset and cross-architecture drift results: two to three months from approval. The updated Portability Risk Index and MLSys 2027 paper follow as downstream outputs on their own timeline.